Taimur Karim

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.07
Holy Angels' School, Katwa Purba Bardhaman, West Bengal	Class XII (ISC)	2023	93.25%
Holy Angels' School, Katwa Purba Bardhaman, West Bengal	Class X (ICSE)	2021	96.00%

SCHOLASTIC ACHIEVEMENTS

Awarded the Academic Excellence Award during the second year for achieving a 9.43 CGPA.

WORK EXPERIENCE

iOS Mobile Engineering Intern at Expedia Group

- **Test Automation Framework:** Engineered an end-to-end iOS UI test automation framework for the Sort & Filter module across BEX and VRBO brands utilizing Appium 2, XCUITest, WebdriverIO v8, and Cucumber BDD.
- **CI/CD Integration:** Automated 32 scenarios (25 BEX, 7 VRBO) and integrated the test suite into GitHub Actions with comprehensive Allure reporting and Slack notifications whenever any sort & filter component changes are pushed.
- **Performance Optimization:** Reduced CI execution time by 66% (from ~131 mins to ~45 mins) by iterating through sequential, parallel and finally implementing sharded parallel runners.
- **AI Tooling & Infrastructure:** Created an automated AI-assisted local setup tool leveraging Claude Code to eliminate manual environment configuration, auto-diagnose 24+ error patterns, and ensure consistent execution across machines.
- **Automated PR Gatekeeping:** Engineered the CI workflow to act as an automatic safety net for pull requests to the dev branch. Eliminated manual QA requirements by strictly blocking code merges until all test scenarios pass on iOS devices.

PROJECTS

Agentic AI Automated Program Repair System

- Architected an agentic AI repair orchestrator in Python/FastAPI, integrating an LLM-driven patch generator with spectrum-based fault localization. Engineered statistical scoring to rank code suspiciousness and candidate patches, reducing the repository search space before autonomously executing iterative test-validation loops to verify candidate fixes.
- Benchmarked the agentic system against a baseline LLM search agent across 40 reproducible bugs, achieving a 26.67% improvement in test-passing repair rates while reducing autonomous tool calls by 22.22%.

Distributed Expense Settlement Engine

- Architected the C++ core for a scalable expense tracking platform, engineering an O(N log N) Max-Heap and disjoint set union debt simplification algorithm to efficiently resolve simulated graphs of 100,000+ multi-party settlements.
- Designed a PostgreSQL double-entry ledger utilizing Optimistic Concurrency Control to eliminate race conditions during concurrent balance updates. Integrated an event-driven message queue and Redis cache architecture, strictly enforcing ACID properties and guaranteeing zero data loss while achieving a load-tested throughput of 2,000+ TPS at sub-50ms latency.

Generative AI Molecular Discovery

- Engineered a PyTorch-based generative AI pipeline utilizing Conditional Variational Autoencoders (CVAEs) with cyclic beta annealing to effectively resolve posterior collapse, generating 4 novel monomers with similar target properties.
- Implemented a Retrieval-Augmented Generation (RAG) architecture using ChromaDB and a locally hosted DeepSeek-R1 LLM via Ollama to autonomously design Phthalonitrile monomers from a curated 59-monomer dataset based on target thermal properties, identifying 11 additional potential monomers with comparable property profiles.

Product Analytics & Experimentation Platform

- Engineered a relational database architecture using PostgreSQL to ingest and model event-level telemetry for a simulated mobile commerce application, writing advanced SQL pipelines (CTEs, window functions) to execute funnel, cohort, and A/B testing analyses.
- Integrated an LLM-based Analytics Copilot to translate natural language queries into executable SQL for automated product insights, containerizing the complete database and analytics ecosystem using Docker for seamless deployment.

RELEVANT COURSES

Maths Courses: Calculus(I and II), Differential Equations I, Elementary Linear Algebra, Probability, Ordinary Differential Equations, Numerical Methods to Solve ODEs, Multivariable Calculus, Linear Algebra

Programming and Machine Learning courses: Introduction to Programming in C, Introduction to Artificial Intelligence, Discrete Maths for Computer Science, Theory of Computation, Machine learning in Chemistry, Computer Architecture, Operating Systems I

SKILLS

Languages: C++, Java, Python, HTML, CSS, SQL

ML frameworks: scikit-learn, Pytorch, OpenAI Gym

Proficient knowledge of **Data Structures and Algorithms**

Solved 500+ DSA problems across major coding platforms (like Leetcode, codeforces, codechef)

Python libraries: numpy, pandas, matplotlib, seaborn, scipy, scikit-learn

Developer Tools: Git, GitHub, Linux, CI/CD (GitHub Actions), Appium, XCUITest, WebdriverIO

CS Fundamentals: Experience with core principles of DBMS, Computer Networks, Operating Systems, and System Design

Soft Skills: Communication, Pitching, Event Organizing

POSITIONS OF RESPONSIBILITY

Infero (The Algorithmic and Logical Reasoning club) core

Robotix (The robotics club): Programming and ML core

Blockspace (The Blockchain and Crypto club) core

OCS Placement and Internship Coordinator

Class Representative for Industrial Chemistry '23

EXTRACURRICULAR

Hackathons: Participated in several hackathons organized by firms like Adobe, Meltwater, KLA etc.

Sports: Served as the weightlifting coordinator and lead the IITH weightlifting team in the Inter IIT sports meet 2024

NSS: Volunteered in National Service Scheme Events